NASA SPACE APPS CHALLENGE 2026

EcoCell Earth Intelligence

What changes on Earth, where, how much – and is it significant?

Team Ecocell

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Challenge: Be An Earth System Trend Detective! · Angola



THE PROBLEM

Decades of NASA data. Hard to read.

24

years of satellite records, from 2002 to 2026

10

Earth variables from NASA missions and models

64,800

blocks of 1° (about 110 km), each tested for change

Most people can’t tell what changed, where, by how much – or whether it is real. EcoCell answers in a few clicks.

OUR SOLUTION

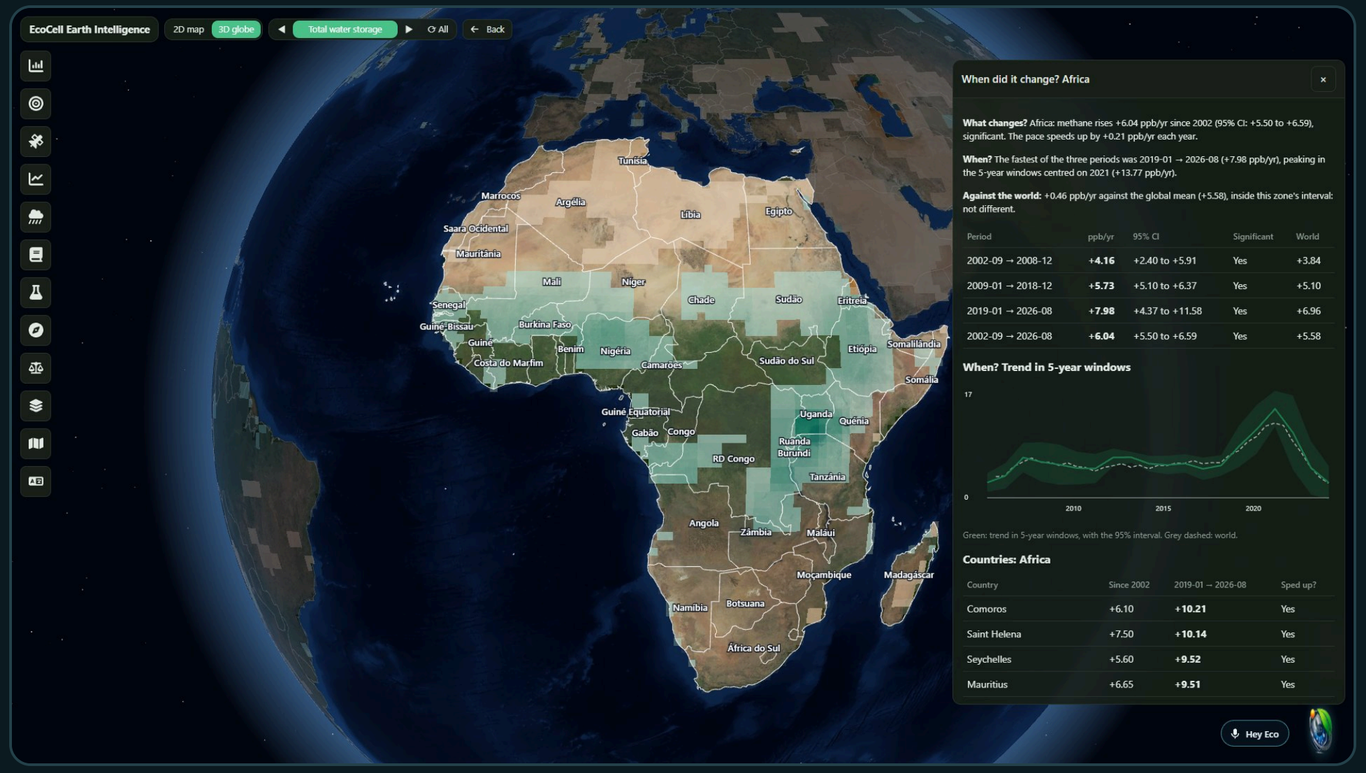
Pick a variable, click a place, get the answer

1 **Choose a NASA variable**
Methane, rainfall, vegetation, water, temperature...

2 **Click a zone**
The world, a continent, a country or one block, on a 2D map or a 3D globe.

3 **See when it changed**
Trend per period, 95% interval, significance, and the world for comparison.

4 **Ask Eco**
Say “Hey Eco” and ask in English or Portuguese.



NASA DATA

Built only on NASA open data

Variable	Mission or model	Years
Methane	Aqua AIRS	2002–2026
Rainfall	GPM IMERG	2003–2024
Vegetation (NDVI)	Terra MODIS	2003–2025
Evapotranspiration	Terra MODIS	2003–2025
Land temperature	Terra MODIS	2003–2020

Variable	Mission or model	Years
Water storage	GRACE, GRACE-FO	2003–2024
Soil moisture	FLDAS model	2003–2025
Air temperature	FLDAS model	2003–2025
Ocean temperature	Aqua MODIS	2003–2025
Ocean chlorophyll	Aqua MODIS	2003–2025

Updated automatically every night from NASA Earthdata and Google Earth Engine, on NASA Blue Marble imagery.

A RESULT: METHANE, NASA AQUA AIRS

Methane is rising – and speeding up

+5.6

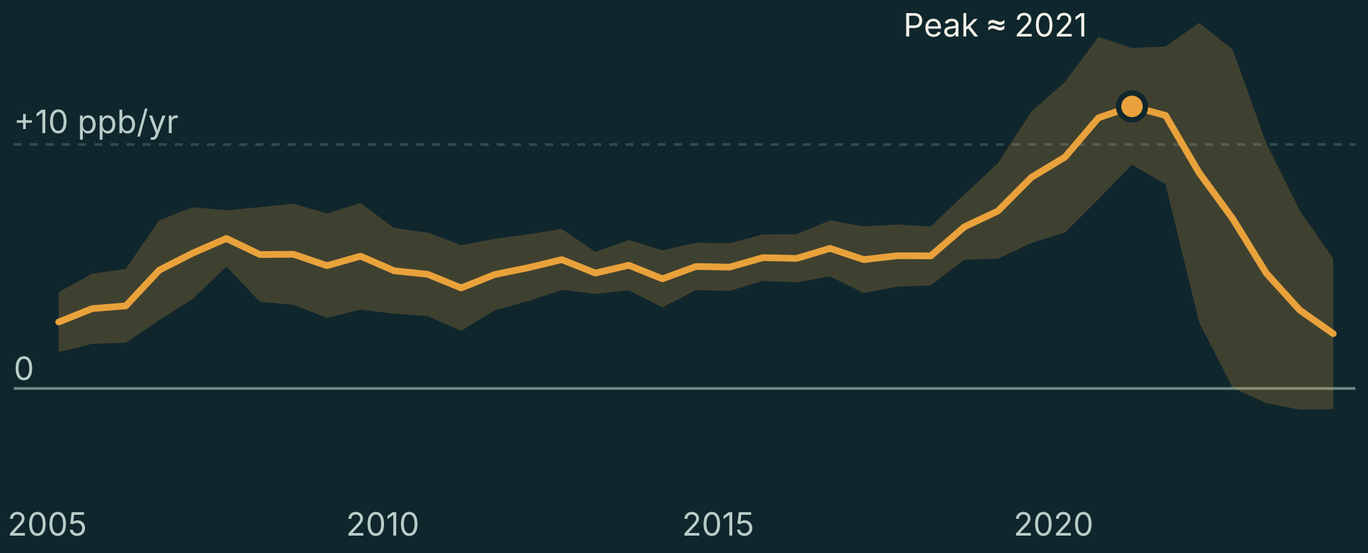
ppb per year worldwide since 2002 (95% CI +4.9 to +6.2)

+0.18 ppb/yr each year

the rise is accelerating (95% CI +0.04 to +0.32)

Fastest around 2021

about +11.6 ppb/yr in 5-year windows



Global trend in 5-year windows, with the 95% interval shaded. Methods: Sen slope, Mann-Kendall test corrected for autocorrelation, false discovery rate q < 0.05. Regional differences that a second instrument did not confirm are shown only as indicative.

BEYOND THE MAP

A voice guide, and a path to action

ECO, THE VOICE GUIDE

Say “Hey Eco”

- Map commands run instantly: “zoom to Brazil”, “show rainfall”, “switch to 3D”.
- Open questions are answered by AI from the site’s live NASA results.
- A guided tour in English and Portuguese.

APPLICATION EXAMPLE

EcoCell: from methane to clean energy

- Methane-eating bacteria in a microbial fuel cell make electricity.
- The electricity splits water into hydrogen, then ammonia.
- The map shows where to study it; a digital twin explains how it works.

A concept: its performance is not yet validated in the lab.

IMPACT AND NEXT STEPS

From NASA data to local decisions

Inform decisions

Show communities and governments where and when their environment changed, with honest uncertainty.

Find methane sources

Add NASA EMIT and future methane missions to locate individual emitters.

Test EcoCell

Build the lab prototype and replace simulated values with measurements.

Try it, free

albertokialamakuntima-glitch.github.io/ecocell

Open source

github.com/albertokialamakuntima-glitch/ecocell

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